

Teknologi Server
Progress Mingguan



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PROGRAM STUDI D-IV TEKNOLOGI REKAYASA INTERNET
DEPARTEMEN TEKNIK ELEKTRO DAN INFORMATIKA
SEKOLAH VOKASI
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1. Bukti Instalasi Nginx dan dashboard Grafana

```
bocahnabine@serverluis:~/hotel-k8s$ sudo apt install nginx -y
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap nginx-doc ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 18
  Download size: 652 kB
  Space needed: 1,853 kB / 2,543 MB available

Get:1 http://id.archive.ubuntu.com/ubuntu/resolute/main amd64 nginx-common all 1.28.3-2ubuntu1 [36.4 kB]
Get:2 http://id.archive.ubuntu.com/ubuntu/resolute/main amd64 nginx amd64 1.28.3-2ubuntu1 [616 kB]
Fetched 652 kB in 3s (255 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database... 98738 files and directories currently installed.)
Preparing to unpack ../nginx-common_1.28.3-2ubuntu1_all.deb...
Unpacking nginx-common (1.28.3-2ubuntu1)...
Selecting previously unselected package nginx.
Preparing to unpack ../nginx_1.28.3-2ubuntu1_amd64.deb...
Unpacking nginx (1.28.3-2ubuntu1)...
Setting up nginx-common (1.28.3-2ubuntu1)...
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1)...
  * Upgrading binary nginx [ OK ]
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for ufw (0.36.2-9build1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```
bocahnabine@serverluis:~/hotel-k8s$ cat grafana.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: grafana-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: grafana
  template:
    metadata:
      labels:
        app: grafana
    spec:
      containers:
        - name: grafana
          image: grafana/grafana:latest
          ports:
            - containerPort: 3000
---
apiVersion: v1
kind: Service
metadata:
  name: grafana-service
spec:
  type: NodePort
  selector:
    app: grafana
  ports:
    - port: 3000
      targetPort: 3000
      nodePort: 32000
```

Proses instalasi web server Nginx pada sistem operasi Ubuntu menggunakan manajer paket apt berjalan dengan lancar. Seluruh paket utama Nginx beserta dependensinya telah berhasil diunduh dan dipasang ke dalam sistem

2. Deploy grafana.yaml

```
bocahnbine@serverluis:~/hotel-k8s$ sudo kubectl apply -f grafana.yaml
deployment.apps/grafana-deployment unchanged
service/grafana-service unchanged
```

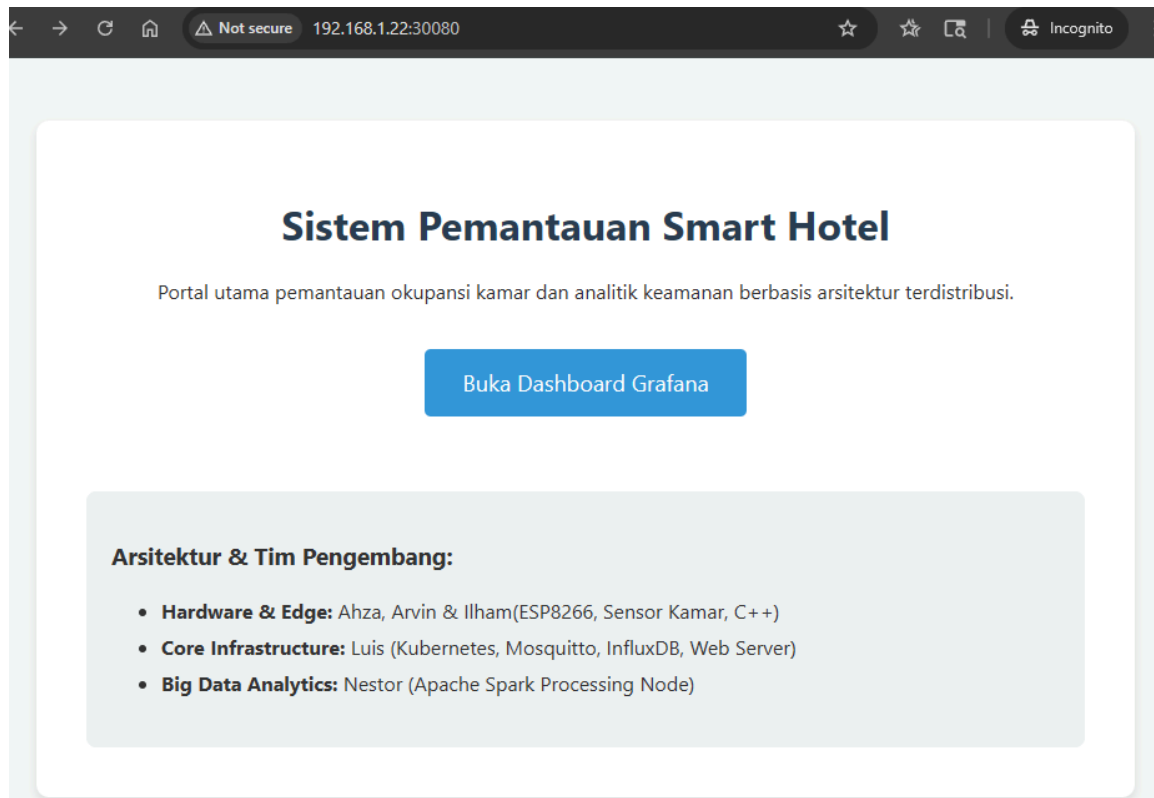
Eksekusi file konfigurasi Grafana menggunakan perintah kubectl apply telah berhasil diterapkan ke dalam kluster.

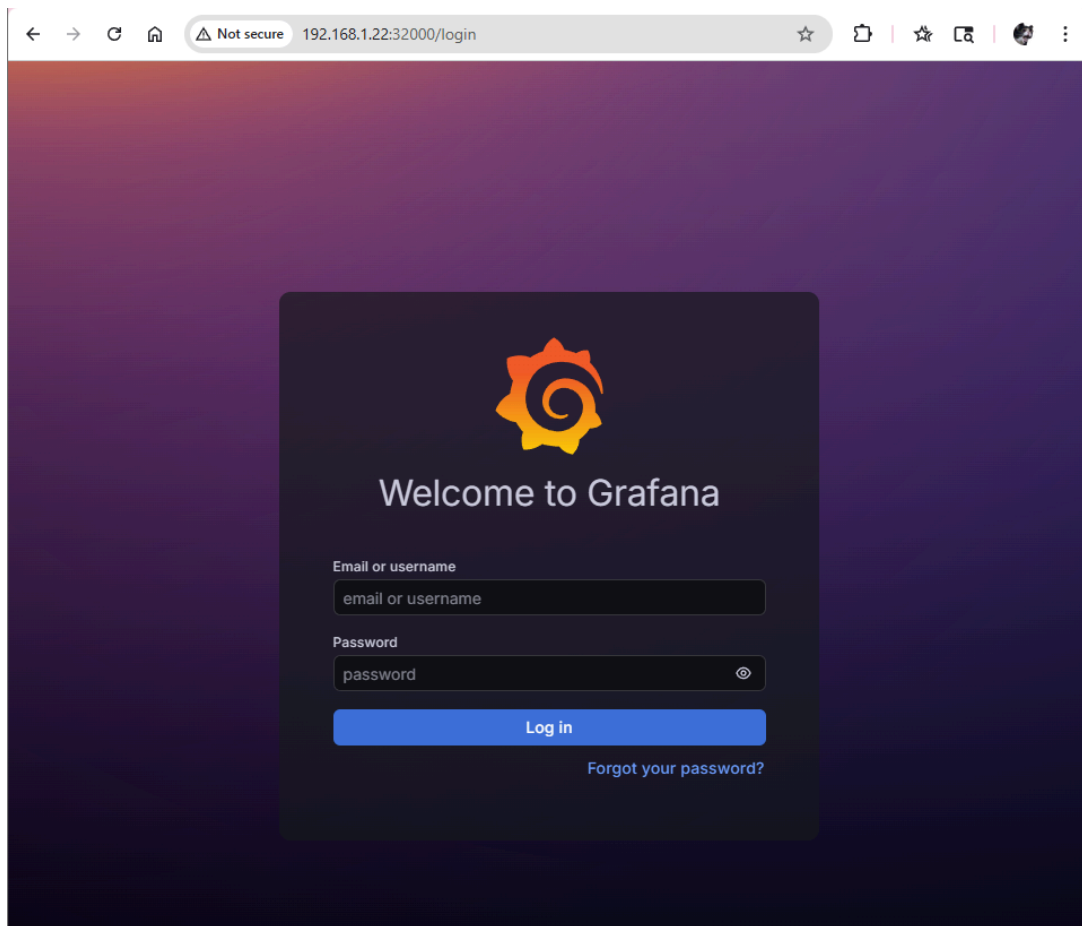
3. Cek Status Pod/Container

```
bocahnbine@serverluis:~/hotel-k8s$ sudo kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
grafana-deployment-567857c7d-px26f  1/1     Running   8 (12m ago)  14d
influxdb-deployment-6b5c679488-rdwzq  1/1     Running   9 (12m ago)  14d
mosquitto-deployment-77fc6b7894-pstmx  1/1     Running   8 (12m ago)  14d
nginx-deployment-56496c457d-w79lx     1/1     Running   6             9d
```

Daftar pods dari arsitektur sistem hotel seluruhnya sudah berstatus Running.

4. Bukti grafana dapat diakses melalui web server





Sistem berhasil merespons panggilan menggunakan alamat IP server beserta port eksternalnya masing-masing.

5. Validasi dan Konfigurasi Nginx + Reverse Proxy

```
bocahnbine@serverluis:~/hotel-k8s$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

Hal ini memvalidasi bahwa script reverse proxy Nginx telah ditulis dengan benar dan siap merutekan lalu lintas jaringan tanpa error.